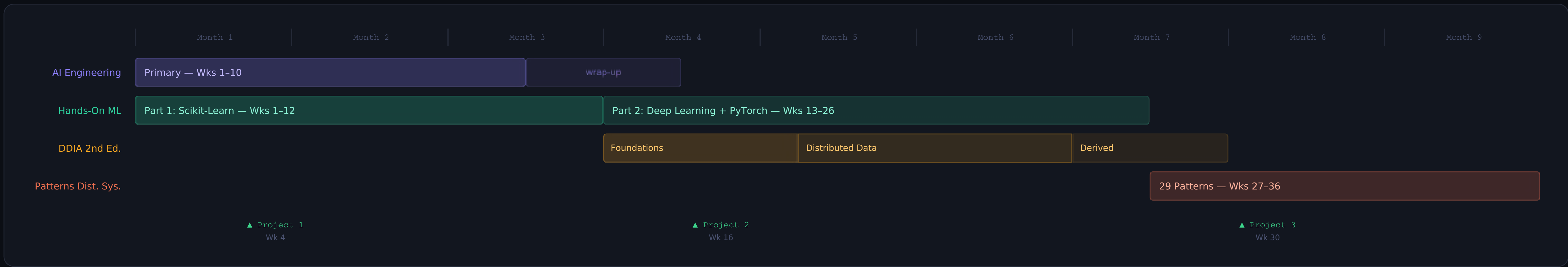
// AI ENGINEER LEARNING ROADMAP

From [illegible]er to AI [illegible]ert

A structured 9-[illegible]s targeting AI eng[illegible].

36 WEEKS TOTAL | 4 BOOKS [illegible]

01 — 9-Month Overview Timeline



02 — Pacing at 10–12 Hours / Week

10 wks
AI Engineering
~50 pg/week · read + reflect

26 wks
Hands-On ML
878 pages · code along

16 wks
DDIA 2nd Edition
~600 pages · think + design

10 wks

Patterns Dist. Sys.

29 patterns · read code

03 — Book-by-Book Plans

BOOK 01 · WEEKS 1-14

AI Engineering: Building Applications with Foundation Models

Chip Huyen — O'Reilly 2025

10 chapters·536 pages·No code, read + build

- Wk 1-3 Foundations — landscape, foundation models, evaluation methodology
- Wk 4-6 Build layer — evaluate AI systems, prompt engineering, RAG + agents ★ Build eval harness + RAG pipeline
- Wk 7-9 Optimise — finetuning, dataset engineering, inference optimisation
- Wk 10-14 Architecture, user feedback + production systems. Wrap-up + project.

BOOK 02 · WEEKS 1-26

Hands-On ML with Scikit-Learn and PyTorch

Aurélien Géron — O'Reilly 2025

19 chapters·878 pages·Code along every chapter

- Wk 1-9 Part 1 — ML landscape, end-to-end project, classification, training models, SVMs, trees, ensembles, dimensionality reduction, unsupervised
- Wk 9 ★ Project 1 — full Scikit-Learn pipeline on Kaggle dataset
- Wk 10-16 Part 2 — PyTorch fundamentals, deep networks, custom models, data loading, CNNs, NLP + transformers (Hugging Face)
- Wk 17-21 Advanced — autoencoders, GANs, diffusion models, RL, deployment at scale
- Wk 22-26 ★ Capstone — end-to-end deep learning project with deployment

BOOK 03 · WEEKS 13-28

Designing Data-Intensive Applications 2nd Edition

Martin Kleppmann + Chris Riccomini — O'Reilly 2026

13 chapters·~620 pages·Read + think + design

- Wk 13-17 Part I Foundations — trade-offs, nonfunctional requirements (new), data models, storage/retrieval, encoding + evolution
- Wk 18-24 Part II Distributed — replication, partitioning, transactions (2 wks), distributed systems trouble, consistency + consensus (2 wks)
- Wk 25-27 Part III Derived — batch processing, stream processing, future of data systems
- Wk 28 Connect to AI engineering — feature stores, pipelines, serving

BOOK 04 · WEEKS 27-36

Patterns of Distributed Systems

Unmesh Joshi — Addison-Wesley 2023

29 patterns · 32 chapters · Java code for every pattern

Wk 27	Narratives — promise/perils overview, pattern dependency map (read Ch 2 twice)
Wk 28–29	Storage + Replication — WAL, Segmented Log, Low-Water Mark, Leader/Followers, HeartBeat, Majority Quorum, Generation Clock, High-Water Mark
Wk 30–32	Consensus + Communication — Paxos, Replicated Log, Singular Update Queue, Idempotent Receiver, Follower Reads, Versioned Value, Version Vector, Fixed/Key-Range Partitions
Wk 33–35	Time + Cluster + Network — Two-Phase Commit, Lamport/Hybrid Clock, Clock-Bound Wait, Consistent Core, Lease, State Watch, Gossip, Emergent Leader, Single-Socket Channel, Batching, Pipelining
Wk 36	★ Final synthesis — connect all 4 books

04 — 3 Portfolio Projects

01

RAG Pipeline + Eval Harness

Build an end-to-end RAG application: ingest documents, embed + store in a vector DB, query with an LLM. Attach a systematic evaluation harness that uses an LLM-as-judge to score outputs across 20+ test cases.

Week 4–6 · AI Engineering + Géron foundations

02

Fine-tuned Deep Learning Model

Pick a real problem (image classification, text classification, or tabular prediction). Fine-tune a pretrained model from Hugging Face. Track experiments with W&B. Deploy to Hugging Face Spaces. Full README as portfolio piece.

Week 14–16 · Géron deep learning chapters

03

AI Data Architecture Design

Design a production AI system architecture: feature store, training pipeline, model serving layer, streaming inference, monitoring. For each component, map the DDIA concept + Joshi pattern that underlies it. Present as a technical design doc.

Week 30–32 · DDIA + Patterns synthesis

05 — How the Four Books Connect

AI Engineering concern

- RAG + retrieval →
- Inference optimisation →
- Fine-tuning →
- Evaluation pipelines →
- Model serving →

Géron ML foundation

- Embedding + nearest neighbour search
- Quantisation, distillation (Ch 19)

Transfer learning (Ch 11, 15)
Metrics: precision, recall, ROUGE
PyTorch, TorchServe (Ch 19)

DDIA systems layer

Storage + indexing (Ch 4)
Partitioning + replication (Ch 6–7)
Encoding + schema evolution (Ch 5)
Stream processing (Ch 12)
Batch processing pipelines (Ch 11)

Joshi implementation

Write-Ahead Log, Segmented Log
Fixed/Key-Range Partitions
Versioned Value, Version Vector
Replicated Log, Consistent Core
Request Batch, Pipeline

06 — Complete 36-Week Schedule

WEEK	BOOK	CHAPTER / FOCUS	KEY PRACTICE TASK	TYPE
Wk 1	AI Engineering	Ch 1 — The AI engineering landscape	Sign up for an LLM API, explore 10+ prompts manually	orient
Wk 1	Hands-On ML	Ch 1 — The ML landscape	Set up Python environment, run first Colab notebook	setup
Wk 2	AI Engineering	Ch 2 — Understanding foundation models	Experiment with temperature + sampling params	deep read
Wk 2	Hands-On ML	Ch 2 — End-to-end ML project (heaviest chapter)	Full California housing pipeline — every step manually	heavy
Wk 3	Hands-On ML	Ch 3 — Classification	Binary + multiclass MNIST classifier, ROC/AUC exploration	code along
Wk 3	AI Engineering	Ch 3 — Evaluation methodology	Design eval rubric for your Week 1 prompts	reflect
Wk 4	AI Engineering	Ch 4 — Evaluate AI systems	★ Build eval harness with LLM-as-judge (20 test cases)	build
Wk 4	Hands-On ML	Ch 4 — Training models (gradient descent, regularisation)	Implement linear regression with NumPy from scratch first	maths heavy
Wk 5	Hands-On ML	Ch 5-6 — SVMs + Decision Trees + XGBoost	SVM on non-linear data, decision tree visualisation	code along
Wk 5	AI Engineering	Ch 5 — Prompt engineering	Compare 3 prompt variants in eval harness, measure scores	build
Wk 6	Hands-On ML	Ch 7 — Ensemble learning + Random Forests	Stacking ensemble + XGBoost on housing dataset	code along
Wk 6	AI Engineering	Ch 6 — RAG and agents	★ Build RAG pipeline with ChromaDB/FAISS, run eval harness on it	build
Wk 7	Hands-On ML	Ch 8 — Dimensionality reduction (PCA, t-SNE)	PCA on MNIST → 2D visualisation, compare t-SNE	code along
Wk 7	AI Engineering	Ch 7 — Finetuning	Write personal fine-tune vs RAG vs prompt decision framework	reflect
Wk 8	Hands-On ML	Ch 9 — Unsupervised learning (K-Means, GMMs)	K-Means on real dataset, elbow + silhouette score	code along
Wk 8	AI Engineering	Ch 8 — Dataset engineering	Audit quality of RAG documents, write data quality report	reflect

Wk 9	Hands-On ML	Part 1 consolidation + ★ Project 1	★ Full Scikit-Learn pipeline on Kaggle dataset → portfolio	★ project
Wk 9	AI Engineering	Ch 9 — Inference optimisation	Benchmark RAG pipeline: latency, tokens/sec, cost per query	build
Wk 10	AI Engineering	Ch 10 — Architecture + user feedback	Sketch production architecture of RAG app end-to-end	design
Wk 10	Hands-On ML	Ch 10 — PyTorch fundamentals (big transition)	Implement linear regression 3 ways: NumPy → tensors → nn.Module	transition
Wk 11	Hands-On ML	Ch 11 — Training deep neural networks	Deep MLP on MNIST, then transfer learning to new dataset	heavy
Wk 12	Hands-On ML	Ch 12–13 — Custom models + data pipelines + HuggingFace	Write custom training loop from scratch; switch to HF Datasets	code along
Wk 13	Hands-On ML	Ch 14 — CNNs + Computer Vision	Train CNN on CIFAR-10; fine-tune ResNet	code along
Wk 13	DDIA 2nd Ed.	Ch 1 — Trade-offs in data systems architecture (new)	Write trade-off analysis of a system you've built at work	reflect
Wk 14	Hands-On ML	Ch 14 exercises + ★ vision project	★ Fine-tune pretrained model on custom image dataset → portfolio	★ project
Wk 14	DDIA 2nd Ed.	Ch 2 — Defining nonfunctional requirements (new)	Write NFR doc for a system: p99 latency, fault rate, load growth	design
Wk 15	Hands-On ML	Ch 15 — NLP + Transformers + Hugging Face (pivotal)	Fine-tune BERT for text classification end-to-end	heavy
Wk 15	DDIA 2nd Ed.	Ch 3 — Data models + query languages	Design same schema 3 ways: relational, document, graph	design
Wk 16	Hands-On ML	Advanced NLP + sequence models + exercises	Fine-tune T5/BART for summarisation, evaluate with ROUGE	code along
Wk 16	DDIA 2nd Ed.	Ch 4 — Storage and retrieval (dense — go slow)	Trace a read through B-tree vs LSM-tree step by step	dense
Wk 17	Hands-On ML	Ch 17 — Autoencoders + GANs	Train VAE on MNIST, visualise latent space sampling	code along
Wk 17	DDIA 2nd Ed.	Ch 5 — Encoding + evolution	Map a past schema incident to Kleppmann's framework	reflect
Wk 18	Hands-On ML	Ch 17 — Diffusion models (most current topic)	HuggingFace Diffusers: generate images + guided generation	cutting edge
Wk 18	DDIA 2nd Ed.	Ch 6 — Replication part 1 (leaders, lag, failover)	Draw replication topology of a DB you use, label every arrow	design
Wk 19	Hands-On ML	Ch 18 — Reinforcement learning + Gymnasium	Train DQN agent to solve CartPole	code along
Wk 19	DDIA 2nd Ed.	Ch 6 cont. + Ch 7 — Partitioning	Design partitioning strategy for 100M ML training examples	design
Wk 20	Hands-On ML	Ch 19 — Training + deploying at scale	Quantise a model, measure size reduction + inference speedup	build
Wk 20	DDIA 2nd Ed.	Ch 8 — Transactions part 1 (ACID + weak isolation)	Build race condition reference card — 4 anomalies with examples	hardest
Wk 21	Hands-On ML	Exercises review + consolidation	Complete all outstanding exercises from deep learning chapters	consolidate
Wk 21	DDIA 2nd Ed.	Ch 8 — Transactions part 2 (serializability + 2PC)	Look up default isolation level of your main DB — is it what you assumed?	dense
Wk 22	Hands-On ML	★ Capstone wk 1 — design + data + baseline	Choose problem, acquire data, run Scikit-Learn baseline	★ project
Wk 22	DDIA 2nd Ed.	Ch 9 — The trouble with distributed systems (revised)	Find a real clock-skew incident post-mortem, map to framework	reflect
Wk 23	Hands-On ML	★ Capstone wk 2 — model + training	Run 5+ training runs, log with W&B, beat baseline	★ project
Wk 23	DDIA 2nd Ed.	Ch 10 — Consistency + consensus part 1 (★ hardest)	Read once, write half-page summary without looking at book	★ hardest
Wk 24	Hands-On ML	★ Capstone wk 3 — evaluation + deployment	Error analysis, deploy to HF Spaces, write README	★ project
Wk 24	DDIA 2nd Ed.	Ch 10 — Consistency + consensus	Watch raft.github.io visualisation, map to etcd/Kubernetes	dense

		part 2 (Raft/Paxos)		
Wk 25	Hands-On ML	Buffer + weak spots review	Pick one unclear concept, go deep: paper + video + code	flex
Wk 25	DDIA 2nd Ed.	Ch 11 — Batch processing (MapReduce + Spark)	Map an ML training pipeline to Kleppmann's batch framework	read
Wk 26	Hands-On ML	Reflect + plan what's next	Write learning retrospective + update 3-month forward plan	done ✓
Wk 26	DDIA 2nd Ed.	Ch 12 — Stream processing (Kafka + real-time)	Design real-time ML feature pipeline with streaming	design
Wk 27	DDIA 2nd Ed.	Ch 13 — The future of data systems	Write 1-page data architecture review of a system you know	synthesis
Wk 27	Patterns	Ch 1–2 — Narratives + Pattern overview map	Draw pattern dependency graph from memory after Ch 2	orient
Wk 28	DDIA 2nd Ed.	Full review + AI engineering synthesis	Build 1-page AI data architecture reference connecting both books	done ✓
Wk 28	Patterns	Ch 3–5 — Write-Ahead Log, Segmented Log, Low-Water Mark	Map WAL → Kafka partition log, Segmented Log → Kafka segments	code + map
Wk 29	Patterns	Ch 6–10 — Leader/Followers, HeartBeat, Majority Quorum, Generation Clock, High-Water Mark	Trace full write request through all 5 patterns as sequence diagram	code along
Wk 30	Patterns	Ch 11–12 — Paxos + Replicated Log (consensus deep dive)	Compare Paxos vs Raft, map Replicated Log → etcd internals	deep
Wk 31	Patterns	Ch 13–16 — Singular Update Queue, Request Waiting List, Idempotent Receiver, Follower Reads	Design idempotent LLM API endpoint with unique ID + response cache	build
Wk 32	Patterns	Ch 17–20 — Versioned Value, Version Vector, Fixed Partitions, Key-Range Partitions	Compare Cassandra (hash) vs CockroachDB (key-range) partitioning	design
Wk 33	Patterns	Ch 21–24 — Two-Phase Commit, Lamport Clock, Hybrid Clock, Clock-Bound Wait	Read Spanner's TrueTime paper, connect to Clock-Bound Wait pattern	time is hard
Wk 34	Patterns	Ch 25–29 — Consistent Core, Lease, State Watch, Gossip, Emergent Leader	Map ZooKeeper: znodes→Consistent Core, ephemeral nodes→Lease, watchers→State Watch	map to systems
Wk 35	Patterns	Ch 30–32 — Single-Socket Channel, Request Batch, Request Pipeline	Map Kafka producer config: linger.ms→Batching, max.in.flight→Pipelining	map + perf
Wk 36	Patterns	★ Final synthesis — four-book integration	★ Build AI systems stack reference: Géron + Huyen + DDIA + Joshi per concern	done ✓